

STANDARD 2

Stage 6

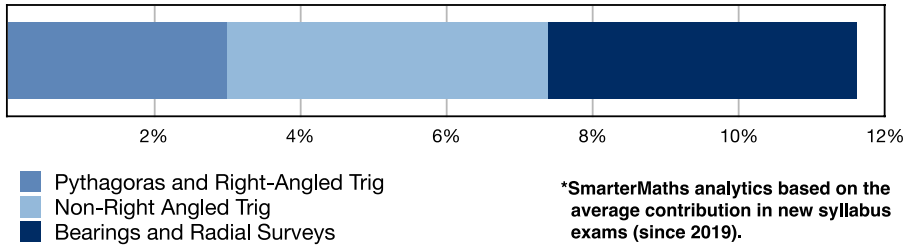
Measurement (Std 2), M6 Trigonometry (Y12)

Pythagoras and Right-Angled Trig (Std2)

Teacher: Hailey Summer

Exam Equivalent Time: 72 minutes (based on allocation of 1.5 minutes per mark)

STD2 Exam Contribution History M6 - Trigonometry



IMPORTANT FEATURES AND TIPS FROM EXAM HISTORY

- MS-M6 Trigonometry* has been a very significant contributor to new syllabus Std2 exams, responsible for an average of 11.6% per exam since 2019.
- We have split this area into 3 subtopics: *Pythagoras and Right-Angled Trig* (3.0%), *Non-Right-Angled Trig* (4.4%) and *Bearings* (4.2%).
- This analysis looks at *Pythagoras and Right-Angled Trig* (3.0%).

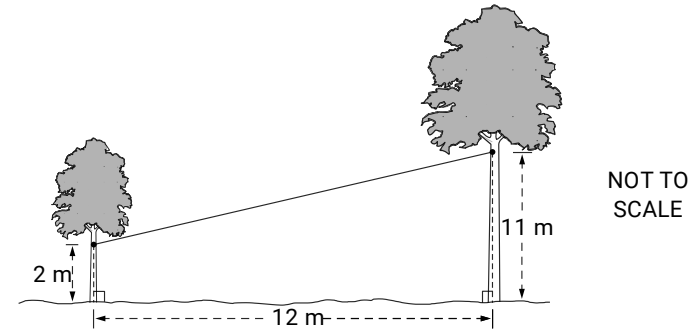
ANALYSIS - What to Expect and Common pitfalls

- Pythagoras and Right-Angled Trig* can be tested via multiple choice or longer answer style questions. It has been examined 7 times in the last decade (most recently in 2023) with 2 separate questions appearing in 2020, 2019 and 2017.
- This sub-topic is a prime area to pick up marks, but note that questions in 2023 (*Q12 MC*), 2020 (*Q32*), 2015 (*Q9 MC*), 2012 (*Q27d*) and 2009 (*Q23a*) produced sub-50% mean marks and deserve attention.
- Rounding an angle to the nearest minute is specifically mentioned in the new syllabus and should be reviewed. We have adjusted "nearest degree" in some past HSC questions to "nearest minute", testing students with some harder rounding examples.

Questions

1. Measurement, STD2 M6 2011 HSC 9 MC

Two trees on level ground, 12 metres apart, are joined by a cable. It is attached 2 metres above the ground to one tree and 11 metres above the ground to the other.

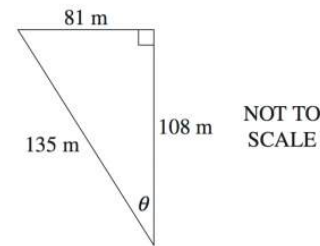


What is the length of the cable between the two trees, correct to the nearest metre?

- A. 9 m
- B. 12 m
- C. 15 m
- D. 16 m

2. Measurement, STD2 M6 2013 HSC 4 MC

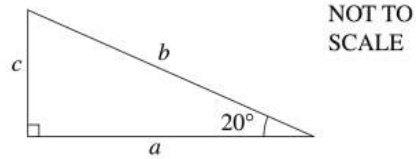
What is the value of θ , to the nearest degree?



- A. 31°
- B. 37°
- C. 49°
- D. 53°

3. Measurement, STD2 M6 2004 HSC 5 MC

What is the correct expression for $\tan 20^\circ$ in this triangle?

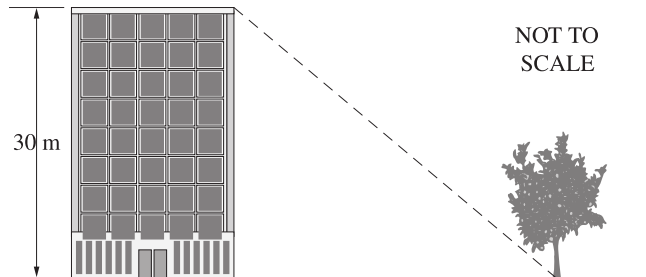


- A. $\frac{a}{b}$
- B. $\frac{a}{c}$
- C. $\frac{c}{b}$
- D. $\frac{c}{a}$

4. Measurement, STD2 M6 2006 HSC 3 MC

The angle of depression of the base of the tree from the top of the building is 65° . The height of the building is 30 m.

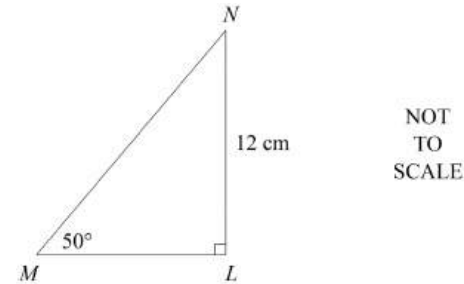
How far away is the base of the tree from the building, correct to one decimal place?



- A. 12.7 m
- B. 14.0 m
- C. 33.1 m
- D. 64.3 m

5. Measurement, STD2 M6 2007 HSC 8 MC

What is the length of the side MN in the following triangle, correct to two decimal places?

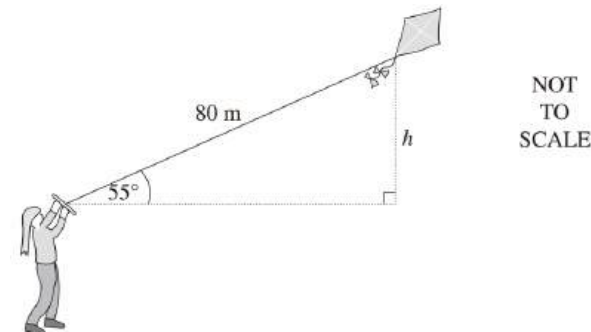


- A. 9.19 cm
- B. 10.07 cm
- C. 15.66 cm
- D. 18.67 cm

6. Measurement, STD2 M6 2008 HSC 14 MC

Danni is flying a kite that is attached to a string of length 80 metres. The string makes an angle of 55° with the horizontal.

How high, to the nearest metre, is the kite above Danni's hand?



- A. 46 m
- B. 66 m
- C. 98 m
- D. 114 m

7. Measurement, STD2 M6 2005 HSC 8 MC

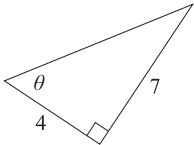
If $\tan\theta = 85$, what is the value of θ , correct to 2 decimal places?

- A. 1.37°
- B. 1.56°
- C. 89.33°
- D. 89.20°

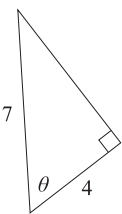
8. Measurement, STD2 M6 SM-Bank 1 MC

In which triangle is $\sin\theta = \frac{4}{7}$?

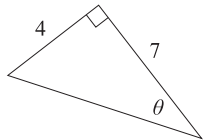
A.



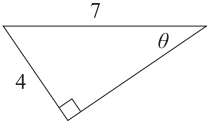
B.



C.

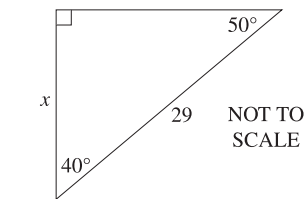


D.



9. Measurement, STD2 M6 2012 HSC 4 MC

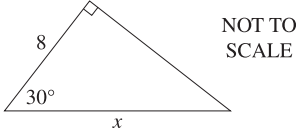
Which expression could be used to calculate the value of x in this triangle?



- A. $29 \times \cos 40^\circ$
- B. $29 \times \cos 50^\circ$
- C. $\frac{\cos 40^\circ}{29}$
- D. $\frac{\cos 50^\circ}{29}$

10. Measurement, STD2 M6 2009 HSC 4 MC

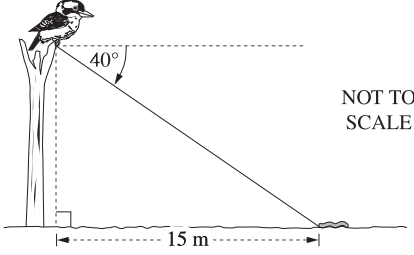
Which is the correct expression for the value of x in this triangle?



- A. $\frac{8}{\cos 30^\circ}$
- B. $\frac{8}{\sin 30^\circ}$
- C. $8 \times \cos 30^\circ$
- D. $8 \times \sin 30^\circ$

11. Measurement, STD2 M6 2011 HSC 4 MC

The angle of depression from a kookaburra's feet to a worm on the ground is 40° . The worm is 15 metres from a point on the ground directly below the kookaburra's feet.

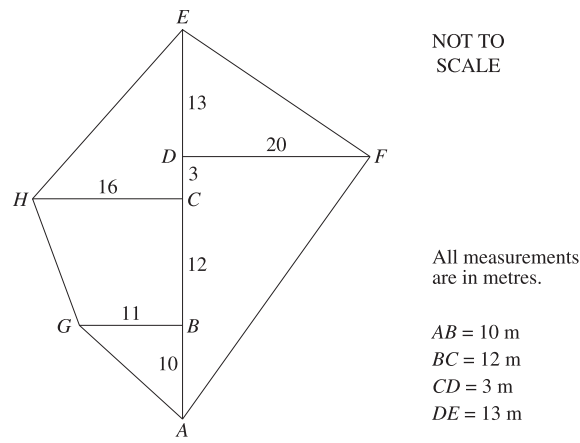


How high above the ground are the kookaburra's feet, correct to the nearest metre?

- A. 10 m
- B. 11 m
- C. 13 m
- D. 18 m

12. Measurement, STD2 M6 2010 HSC 3 MC

A field diagram has been drawn from an offset survey.

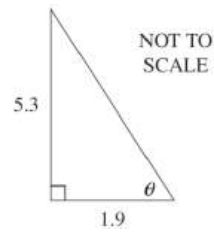


What is the distance from **G** to **H** correct to the nearest metre?

- A. 11
- B. 13
- C. 16
- D. 20

13. Measurement, STD2 M6 2017 HSC 8 MC

The diagram shows a right-angled triangle.

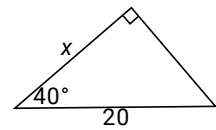


What is the value of θ , to the nearest minute?

- A. $70^{\circ}16'$
- B. $70^{\circ}17'$
- C. $70^{\circ}27'$
- D. $70^{\circ}28'$

14. Measurement, STD2 M6 2024 HSC 6 MC

Consider the diagram shown.

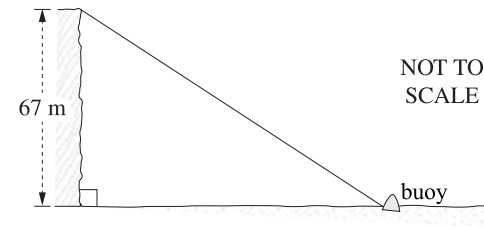


Which of the following is the correct expression for the length of x ?

- A. $20 \cos 40^{\circ}$
- B. $20 \sin 40^{\circ}$
- C. $\frac{20}{\cos 40^{\circ}}$
- D. $\frac{20}{\sin 40^{\circ}}$

15. Measurement, STD2 M6 2015 HSC 9 MC

From the top of a cliff 67 metres above sea level, the angle of depression of a buoy is 42° .



How far is the buoy from the base of the cliff, to the nearest metre?

- A. 60 m
- B. 74 m
- C. 90 m
- D. 100 m

16. Measurement, STD2 M6 2019 HSC 12 MC

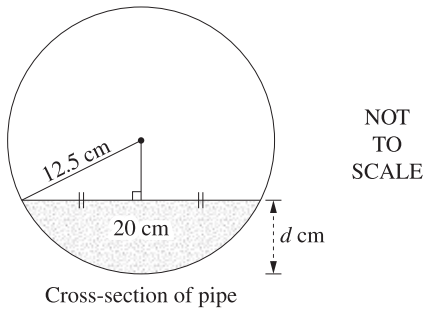
An owl is 7 metres above ground level, in a tree. The owl sees a mouse on the ground at an angle of depression of 32° .

How far must the owl fly in a straight line to catch the mouse, assuming the mouse does not move?

- A. 3.7 m
- B. 5.9 m
- C. 8.3 m
- D. 13.2 m

17. Measurement, STD2 M6 2023 HSC 12 MC

A cylindrical pipe with a radius of 12.5 cm is filled with water to a depth, d cm, as shown.
The surface of the water has a width of 20 cm.

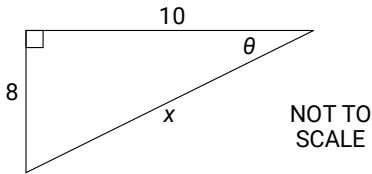


What is the depth of water in the pipe?

- A. 2.5 cm
- B. 5.0 cm
- C. 7.5 cm
- D. 12.5 cm

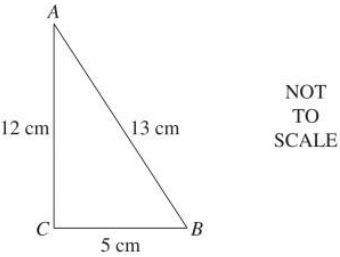
18. Measurement, STD2 M6 2020 HSC 16

Consider the triangle shown.



- a. Find the value of θ , correct to the nearest degree. (2 marks)
- b. Find the value of x , correct to one decimal place. (2 marks)

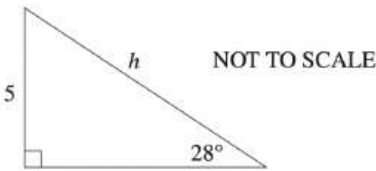
19. Measurement, STD2 M6 2005 HSC 25b



- i. Use Pythagoras' theorem to show that $\triangle ABC$ is a right-angled triangle. (1 mark)
- ii. Calculate the size of $\angle ABC$ to the nearest minute. (2 marks)

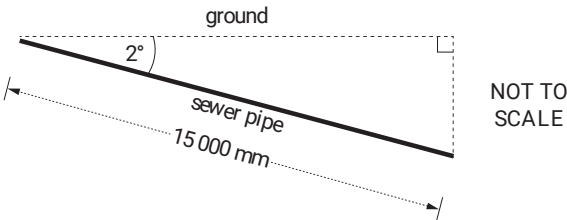
20. Measurement, STD2 M6 2014 HSC 26b

Calculate the value of h correct to two decimal places. (2 marks)



21. Measurement, STD2 M6 2017 HSC 26d

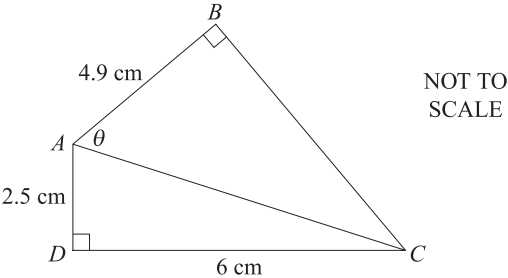
A sewer pipe needs to be placed into the ground so that it has a 2° angle of depression. The length of the pipe is 15 000 mm.



How much deeper should one end of the pipe be compared to the other end? Answer to the nearest mm. (2 marks)

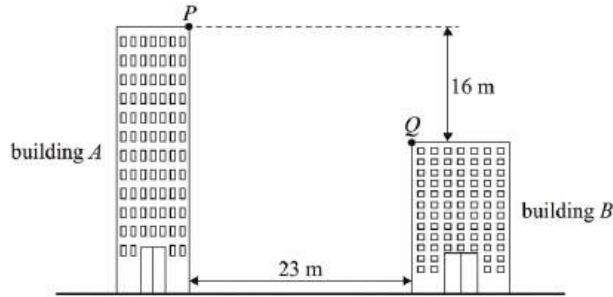
22. Measurement, STD2 M6 2019 HSC 22

Two right-angled triangles, ABC and ADC , are shown.



Calculate the size of angle θ , correct to the nearest minute. (3 marks)

23. Measurement, STD2 M6 FUR1 2011 2



The point Q on building B is visible from the point P on building A , as shown in the diagram above.

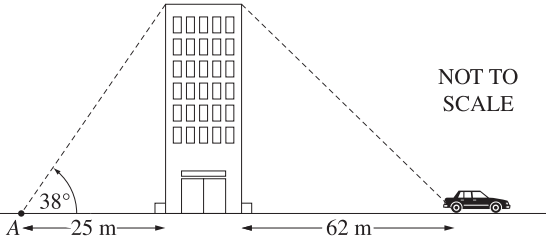
Building A is 16 metres taller than building B .

The horizontal distance between point P and point Q is 23 metres.

Calculate the angle of depression of point Q from point P . Give your answer correct to 1 decimal place. (2 marks)

24. Measurement, STD2 M6 2009 HSC 23a

The point A is 25 m from the base of a building. The angle of elevation from A to the top of the building is 38° .



i. Show that the height of the building is approximately 19.5 m. (1 mark)

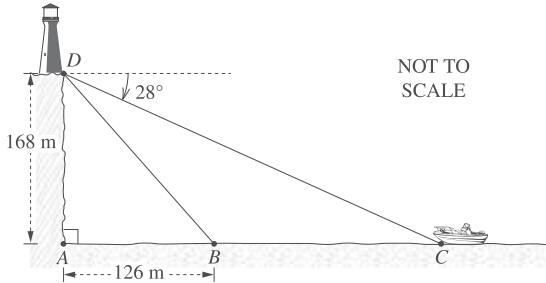
ii. A car is parked 62 m from the base of the building.

What is the angle of depression from the top of the building to the car?

Give your answer to the nearest **minute**. (2 marks)

25. Measurement, STD2 M6 2010 HSC 24d

The base of a lighthouse, D , is at the top of a cliff 168 metres above sea level. The angle of depression from D to a boat at C is 28° . The boat heads towards the base of the cliff, A , and stops at B . The distance AB is 126 metres.



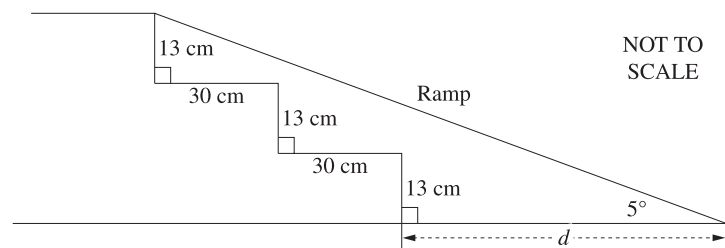
i. What is the angle of depression from D to B , correct to the nearest degree? (3 marks)

ii. How far did the boat travel from C to B , correct to the nearest metre? (2 marks)

26. Measurement, STD2 M6 2012 HSC 27d

A disability ramp is to be constructed to replace steps, as shown in the diagram.

The angle of inclination for the ramp is to be 5° .



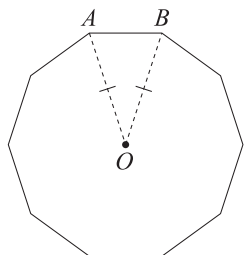
Calculate the extra distance, d , that the ramp will extend beyond the bottom step.

Give your answer to the nearest centimetre. (3 marks)

27. Measurement, STD2 M6 2020 HSC 32

The diagram shows a regular decagon (ten-sided shape with all sides equal and all interior angles equal).

The decagon has centre O .



The perimeter of the shape is 80 cm.

By considering triangle OAB , calculate the area of the ten-sided shape. Give your answer in square centimetres correct to one decimal place. (4 marks)

Worked Solutions

1. Measurement, STD2 M6 2011 HSC 9 MC

Using Pythagoras

$$c^2 = 12^2 + 9^2$$

$$= 144 + 81$$

$$= 225$$

$$\therefore c = 15, \quad c > 0$$

$$\Rightarrow C$$

2. Measurement, STD2 M6 2013 HSC 4 MC

$$\sin \theta = \frac{81}{135}$$

$$\therefore \theta = 37^\circ$$

$$\Rightarrow B$$

Note that $\tan \theta$ or $\cos \theta$ could also be used.

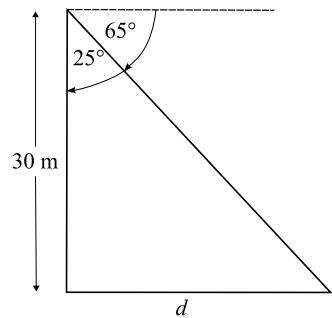
3. Measurement, STD2 M6 2004 HSC 5 MC

$$\tan 20^\circ = \frac{\text{opposite}}{\text{adjacent}}$$

$$= \frac{c}{a}$$

$$\Rightarrow D$$

4. Measurement, STD2 M6 2006 HSC 3 MC



Let d = distance from base to tree

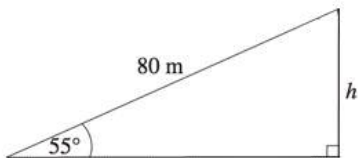
$$\begin{aligned}\tan 25^\circ &= \frac{d}{30} \\ \therefore d &= 30 \times \tan 25^\circ \\ &= 13.98 \dots \text{ m}\end{aligned}$$

$\Rightarrow B$

5. Measurement, STD2 M6 2007 HSC 8 MC

$$\begin{aligned}\sin 50^\circ &= \frac{12}{MN} \\ MN &= \frac{12}{\sin 50^\circ} \\ &= 15.66 \dots \\ \Rightarrow C\end{aligned}$$

6. Measurement, STD2 M6 2008 HSC 14 MC

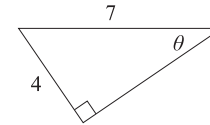


$$\begin{aligned}\sin 55^\circ &= \frac{h}{80} \\ h &= 80 \times \sin 55^\circ \\ &= 65.532 \dots \text{ m} \\ \Rightarrow B\end{aligned}$$

7. Measurement, STD2 M6 2005 HSC 8 MC

$$\begin{aligned}\tan \theta &= 85 \\ \theta &= \tan^{-1} 85 \\ &= 89.33^\circ \\ \Rightarrow C\end{aligned}$$

8. Measurement, STD2 M6 SM-Bank 1 MC



$\Rightarrow D$

9. Measurement, STD2 M6 2012 HSC 4 MC

$$\begin{aligned}\cos 40^\circ &= \frac{x}{29} \\ x &= 29 \times \cos 40^\circ \\ \Rightarrow A\end{aligned}$$

♦ Mean mark 42%

10. Measurement, STD2 M6 2009 HSC 4 MC

$$\begin{aligned}\cos 30^\circ &= \frac{8}{x} \\ \therefore x &= \frac{8}{\cos 30^\circ} \\ \Rightarrow A\end{aligned}$$

11. Measurement, STD2 M6 2011 HSC 4 MC

$$\begin{aligned}\angle \text{Elevation (worm)} &= 40^\circ \quad (\text{alternate angles}) \\ \tan 40^\circ &= \frac{h}{15} \\ \therefore h &= 15 \times \tan 40^\circ \\ &= 12.58 \dots \text{ m} \\ \Rightarrow C\end{aligned}$$

12. Measurement, STD2 M6 2010 HSC 3 MC

Using Pythagoras:

$$\begin{aligned} GH^2 &= 12^2 + (16 - 11)^2 \\ &= 144 + 25 \\ &= 169 \end{aligned}$$

$$\begin{aligned} \therefore GH &= \sqrt{169} \\ &= 13 \text{ m} \end{aligned}$$

$\Rightarrow B$

13. Measurement, STD2 M6 2017 HSC 8 MC

$$\begin{aligned} \tan \theta &= \frac{\text{opp}}{\text{adj}} \\ &= \frac{5.3}{1.9} \\ &= 2.789\dots \end{aligned}$$

$$\begin{aligned} \therefore \theta &= 70.277\dots^\circ \\ &= 70^\circ 16' 39.8'' \\ &= 70^\circ 17' \end{aligned}$$

$\Rightarrow B$

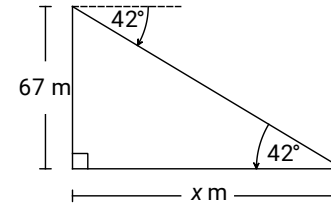
14. Measurement, STD2 M6 2024 HSC 6 MC

$$\begin{aligned} \cos 40^\circ &= \frac{x}{20} \\ x &= 20 \times \cos 40^\circ \end{aligned}$$

$\Rightarrow A$

COMMENT: An angle that has over 30" (seconds) is rounded up to the next minute (i.e. rounded up to 70°17').

15. Measurement, STD2 M6 2015 HSC 9 MC



Let x = distance of buoy from cliff base

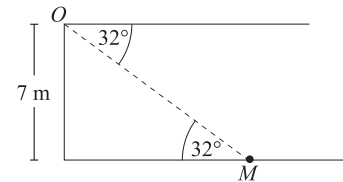
$$\begin{aligned} \tan 42^\circ &= \frac{67}{x} \\ x \tan 42^\circ &= 67 \\ x &= \frac{67}{\tan 42^\circ} \\ &= 74.41\dots \text{ m} \end{aligned}$$

$\Rightarrow B$

♦ Mean mark 49%.

COMMENT: The angle of depression is a regularly examined concept. Make sure you know exactly what it refers to.

16. Measurement, STD2 M6 2019 HSC 12 MC



Let OM = Flight distance

$$\begin{aligned} \sin 32^\circ &= \frac{7}{OM} \\ \therefore OM &= \frac{7}{\sin 32^\circ} \\ &= 13.2 \text{ m} \end{aligned}$$

$\Rightarrow D$

♦ Mean mark 36%.

17. Measurement, STD2 M6 2023 HSC 12 MC

Triangle base $= \frac{20}{2} = 10 \text{ cm}$

Let $x = \perp$ distance from centre to water

Using Pythagoras:

$$\begin{aligned} 12.5^2 &= x^2 + 10^2 \\ x^2 &= 12.5^2 - 10^2 \\ &= 56.25 \\ x &= 7.5 \text{ cm} \end{aligned}$$

$$\begin{aligned} d &= 12.5 - 7.5 = 5.0 \text{ cm} \\ \Rightarrow B \end{aligned}$$

♦ Mean mark 42%.

18. Measurement, STD2 M6 2020 HSC 16

a. $\tan \theta = \frac{8}{10}$

$$\begin{aligned} \theta &= \tan^{-1} \frac{8}{10} \\ &= 38.659\dots \\ &= 39^\circ \text{ (nearest degree)} \end{aligned}$$

b. Using Pythagoras:

$$\begin{aligned} x &= \sqrt{8^2 + 10^2} \\ &= 12.806\dots \\ &= 12.8 \text{ (to 1 d.p.)} \end{aligned}$$

19. Measurement, STD2 M6 2005 HSC 25b

i. $\triangle ABC$ is right-angled if $a^2 + b^2 = c^2$

$$\begin{aligned} a^2 + b^2 &= 5^2 + 12^2 \\ &= 169 \\ &= 13^2 \\ &= c^2 \dots \text{ as required.} \end{aligned}$$

ii. $\sin \angle ABC = \frac{12}{13}$

$$\begin{aligned} \therefore \angle ABC &= 67.38\dots^\circ \\ &= 67^\circ 22' 48'' \\ &= 67^\circ 23' \text{ (nearest minute)} \end{aligned}$$

MARKER'S COMMENT: Know your calculator process for producing an angle in minutes/seconds. Note >30 "seconds" rounds up to the higher "minute".

20. Measurement, STD2 M6 2014 HSC 26b

$$\begin{aligned} \sin 28^\circ &= \frac{5}{h} \\ \therefore h &= \frac{5}{\sin 28^\circ} \\ &= 10.6502\dots \\ &= 10.65 \text{ (2 d.p.)} \end{aligned}$$

21. Measurement, STD2 M6 2017 HSC 26d

Let $x = \text{depth needed}$

$$\begin{aligned} \sin 2^\circ &= \frac{x}{15\,000} \\ x &= 15\,000 \times \sin 2^\circ \\ &= 523.49\dots \\ &= 523 \text{ mm (nearest mm)} \end{aligned}$$

22. Measurement, STD2 M6 2019 HSC 22

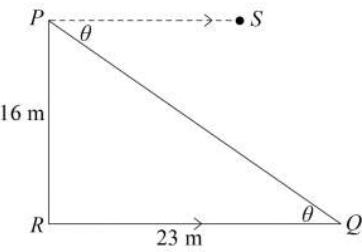
Using Pythagoras in $\triangle ACD$:

$$\begin{aligned} AC^2 &= 2.5^2 + 6^2 \\ &= 42.25 \\ \therefore AC &= 6.5 \text{ cm} \end{aligned}$$

In $\triangle ABC$:

$$\begin{aligned} \cos \theta &= \frac{4.9}{6.5} \\ \theta &= \cos^{-1} \frac{4.9}{6.5} \\ &= 41.075\dots \\ &= 41^\circ 4' 31'' \\ &= 41^\circ 5' \text{ (nearest minute)} \end{aligned}$$

23. Measurement, STD2 M6 FUR1 2011 2



$$\begin{aligned} \tan \theta &= \frac{16}{23} \\ \theta &= 34.8^\circ \end{aligned}$$

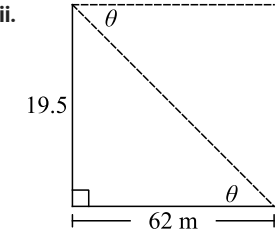
$\therefore$ Angle of depression from P to Q = 34.8°
(alternate, $PS \parallel RQ$)

Mean mark 51%.

24. Measurement, STD2 M6 2009 HSC 23a

i. Need to prove height (h) ≈ 19.5 m

$$\begin{aligned} \tan 38^\circ &= \frac{h}{25} \\ h &= 25 \times \tan 38^\circ \\ &= 19.5321\dots \\ &\approx 19.5 \text{ m} \dots \text{as required.} \end{aligned}$$



Let $\angle$ Elevation (from car) = θ

$$\begin{aligned} \tan \theta &= \frac{h}{62} \\ &= \frac{19.5}{62} \\ &= 0.3145\dots \\ \therefore \theta &= 17.459\dots \\ &= 17^\circ 27' 33'' \dots \\ &= 17^\circ 28' \text{ (nearest minute)} \end{aligned}$$

$\therefore \angle$ Depression to car = $17^\circ 28'$ (alternate to θ)

♦♦ Mean mark 33%
MARKER'S COMMENT: If >30
"seconds", round to the higher
"minute".

25. Measurement, STD2 M6 2010 HSC 24d

i. $\tan \angle ADB = \frac{126}{168}$
 $\angle ADB = 36.8698\dots$
 $= 36.9^\circ \text{ (to 1 d.p)}$

$\angle \text{Depression } D \text{ to } B = 90 - 36.9$
 $= 53.1$
 $= 53^\circ \text{ (nearest degree)}$

ii. Find CB :

$\angle ADC + 28 = 90$
 $\angle ADC = 62^\circ$

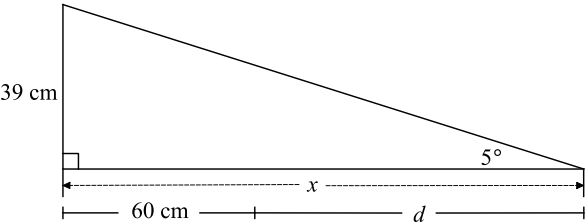
$\tan 62^\circ = \frac{AC}{168}$
 $AC = 168 \times \tan 62^\circ$
 $= 315.962\dots$

$CB = AC - AB$
 $= 315.962\dots - 126$
 $= 189.962\dots$
 $= 190 \text{ m (nearest m)}$

♦♦ Mean mark 31%

♦♦ Mean mark 31%
MARKER'S COMMENT: Solve efficiently by using right-angled trigonometry. Many students used non-right angled trig, adding to the calculations and the difficulty.

26. Measurement, STD2 M6 2012 HSC 27d



Let the horizontal part of the ramp = x cm

$\tan 5^\circ = \frac{39}{x}$
 $x = \frac{39}{\tan 5^\circ}$
 $= 445.772\dots$

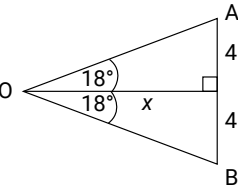
Since $x = 60 + d$
 $d = 445.772 - 60$
 $= 385.772 \text{ cm}$
 $= 386 \text{ cm (nearest cm)}$

♦♦ Mean mark 35%
MARKER'S COMMENT: The better responses used a diagram of a simplified version of the ramp as per the Worked Solution.

$\angle AOB = \frac{360}{10} = 36^\circ$

$AB = \frac{80}{10} = 8 \text{ cm}$

$\triangle AOB$ is made up of 2 identical right-angled triangles



$\tan 18^\circ = \frac{4}{x}$

$x = \frac{4}{\tan 18^\circ}$

$\therefore \text{Area of decagon} = 20 \times \frac{1}{2} \times \frac{4}{\tan 18^\circ} \times 4$
 $= 492.429\dots$
 $= 492.4 \text{ cm}^2 \text{ (to 1 d.p.)}$

♦♦ Mean mark 33%.